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# LOGARITHMS

## NIMCET PYQ

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1. If $a = 1 + \log_x yz, b = 1 + \log_y xz, c = 1 + \log_z xy$, then $ab + bc + ca$ is equal to

**NIMCET 2007**

(a)0 (b)2abc
(c)abc (d)$a^2 + b^2 + c^2$

2. If $0 < y < 2^{1/3}$ and $x(y^3 - 1) = 1$, then $\frac{2}{x} + \frac{2}{3x^3} + \frac{2}{5x^5} + \cdots =$

**NIMCET 2007**

(a)$\log\left[\frac{y^3}{2-y^3}\right]$ (b)$\log\left[\frac{y^3}{1-y^3}\right]$
(c)$\log\left[\frac{2y^3}{1-y^3}\right]$ (d)$\log\left[\frac{y^3}{1-2y^3}\right]$

3. The value of $y = 0.36\log_{0.25}\left(\frac{1}{3} + \frac{1}{3^2} + \cdots\right)$ is

**NIMCET 2008**

(a)0.9 (b)0.18 (c)0.6 (d)0.25

4. For $a > 0, a \neq 1$, the number of values of x satisfying the equation $2\log_x(a) + \log_{ax}(a) + 3\log_{a^2x}(a) = 0$

**NIMCET 2008**

(a)2 (b)3 (c)4 (d)5

5. If $x < -1$ and $2^{|x+1|} - 2^x = |2^x - 1| + 1$, then the value of x is

**NIMCET 2009**

(a)$-2$ (b)2 (c)0 (d)1

6. The number of distinct integral values of 'a' satisfying the equation $2^{2a} - 3(2^{a+2}) + 2^5 = 0$ is

**NIMCET 2009**

(a)0 (b)1 (c)2 (d)3

7. Solution set of the inequality $\log_3(x+2)(x+4) + \log_{1/3}(x+2) < \frac{1}{2}\log_{\sqrt{3}} 7$ is

**NIMCET 2011**

(a)$(-2,-1)$ (b)$(-2,3)$
(c)$(-1,3)$ (d)$(3,\infty)$

8. If $x = \log_a bc, y = \log_b ca$ and $z = \log_c ab$, then $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$ is equal to

**NIMCET 2012**

(a)abc (b)$\sqrt{ab} + \sqrt{bc} + \sqrt{ca}$
(c)1 (d)$x + y + z$

9. If $2^a = 3^b = 6^{-c}$, then $ab + bc + ca$ is equal to

**NIMCET 2012**

(a)1 (b)2 (c)0 (d)None

10. If $x = \log_3 5, y = \log_{17} 25$, then which one of the following is correct?
(a)$x > y$ (b)$x < y$ (c)$x \leq y$ (d)$x = y$

11. If $\log_x y = 100$ and $\log_2 x = 10$, then the value of y is

**NIMCET 2013**

(a)$2^{10}$ (b)$2^{100}$ (c)$2^{1000}$ (d)$2^{10000}$

12. If $(x_0, y_0)$ is the solution of the following equations. $(2x)^{\ln 2} = 3y^{\ln x} = 2^{\ln y}$. Then, $x_0$ is

**NIMCET 2014**

(a)1/6 (b)1/3 (c)1/2 (d)6

13. If $a = \log_{12} 18, b = \log_{24} 54$, then $ab + 5(a - b)$ is

**NIMCET 2014**

(a)1 (b)0 (c)2 (d)3/2

14. If x, y and z are three consecutive positive integers, then $\log(1 + xz)$ is

**NIMCET 2015**

(a)$\log y$ (b)$\log\frac{y}{2}$ (c)$\log(2y)$ (d)$2\log(y)$

15. The solution set of equation $\log_x 2 \log_{2x} 2 = \log_{4x} 2$ is

**NIMCET 2016**

(a)$\left\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\right\}$ (b)$\left\{\frac{1}{2}, 2\right\}$
(c)$\left\{\frac{1}{4}, 2^2\right\}$ (d)$\left\{\frac{1}{4}, 2\right\}$

16. If $3^x = 4^{x-1}$, then x is equal to **NIMCET 2016**

(a)$\frac{2-\log_3 2}{2\log_3 2 - 1}$ (b)$\frac{2}{2\log_3 2 - 1}$
(c)$\frac{2-\log_3 2}{2\log_3 2 + 1}$ (d)$\frac{2\log_3 2}{2\log_3 2 - 1}$

17. Sum of the roots of the equations $4^x - 3(2^{x+3}) + 128 = 0$ is

**NIMCET 2016**

(a)5 (b)6 (c)7 (d)8

18. $6 + \log_{1/4}\frac{1}{\sqrt{2}}\left[\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 - \frac{1}{\sqrt{2}} \ldots \ldots}}}\right]$ is equal to

**NIMCET 2018**

(a)6 (b)13/2 (c)4 (d)25/4

19. Solution set of the inequality $\log_3(x+2)(x+4) + \log_{1/3}(x+2) < \frac{1}{2}\log_{\sqrt{3}} 7$ is

**NIMCET 2019**

(a)$(-2,-1)$ (b)$(-2,3)$
(c)$(-1,3)$ (d)$(3,\infty)$

20. The value of $3^{3-\log_3 5}$ is **NIMCET 2022**

(a)$\frac{5}{27}$ (b)$\frac{27}{5}$ (c)$\frac{9}{5}$ (d)$\frac{5}{9}$

21. Which of the following number is the coefficient of $x^{100}$ in the expansion of $\log_e\left(\frac{1+x}{1+x^2}\right), |x| < 1$?

**NIMCET 2023**

(a)0.01 (b)0.02 (c)$-0.03$ (d)$-0.01$

22. Let x be a positive real number such that

$x^{(8\log_5(x)-24)} = 5^{-4}$.

Then the product of all possible values of x is

**NIMCET 2025**

(a) 125 (b) 625
(c) 25 (d) 5

23. If $8^{x-1} = \left(\frac{1}{4}\right)^x$, then the value of $\frac{1}{\log_{x+1}4 - \log_{x+1}5} + \frac{1}{\log_{1-x}4 - \log_{1-x}5}$ is

**NIMCET 2025**

(a) $\frac{5}{4}$ (b) 1 (c) 2 (d) $\frac{4}{5}$

## ANSWER KEY

| 1. | 2. | 3. | 4. | 5. | 6. | 7. | 8. | 9. | 10. |
|---|---|---|---|---|---|---|---|---|---|
| c | a | b | a | a | c | b | c | c | a |
| 11. | 12. | 13. | 14. | 15. | 16. | 17. | 18. | 19. | 20. |
| c | c | a | d | a | d | c | b | b | b |
| 21. | 22. | 23. | | | | | | | |
| a | a | c | | | | | | | |

## Analysis 2007 to 2025
## Questions Asked in NIMCET 2007 to 2025

| Years | No. of Questions | Years | No. of Questions |
|---|---|---|---|
| NIMCET 2007 | 02 | NIMCET 2016 | 03 |
| NIMCET 2008 | 02 | NIMCET 2017 | 00 |
| NIMCET 2009 | 02 | NIMCET 2018 | 01 |
| NIMCET 2010 | 00 | NIMCET 2019 | 01 |
| NIMCET 2011 | 01 | NIMCET 2020 | 00 |
| NIMCET 2012 | 02 | NIMCET 2021 | 00 |
| NIMCET 2013 | 01 | NIMCET 2022 | 01 |
| NIMCET 2014 | 02 | NIMCET 2023 | 01 |
| NIMCET 2015 | 01 | NIMCET 2024 | 00 |
| | | NIMCET 2025 | 02 |

## SOLUTION

**1. (c)** a = 1 + $\log_x$ yz

b = 1 + $\log_y$ xz

c = 1 + $\log_z$ xy

a = $\log_x$ x + $\log_x$ yz = $\log_x$ xyz

b = $\log_y$ xyz

c = $\log_z$ xyz

ab + bc + ca

($\log_x$ xyz) ($\log_y$xyz) + $\log_y$xyz $\log_z$xyz + $\log_z$xyz $\log_x$xyz

$$\frac{(\log\ xyz)^2}{\log x.\log y} + \frac{(\log xyz)^2}{\log y.\log z} + \frac{(\log(xyz))^2}{\log z.\log y}$$

$$(\log xyz)^2 \left\{\frac{1}{\log x.\log y} + \frac{1}{\log y.\log z} + \frac{1}{\log z\log x}\right\}$$

$$(\log xyz)^2 \left[\frac{\log x+\log y+\log z}{\log x.\log y.\log z}\right]$$

$$\frac{(\log xyz)^3}{\log x\ \log y\ \log z} = \log_x xyz \,.\, \log_y xyz \,.\log_z xyz = abc$$

**2. (a)** $x(y^3-1)=1$ Value of $\frac{2}{x}+\frac{2}{3x^3}+\frac{2}{5x^5}+\cdots\infty$ we know that

$$\log\left(\frac{1+x}{1-x}\right) = 2\left[x+\frac{x^3}{3}+\frac{x^5}{5}+\cdots\infty\right)$$

$$\ln\left[\frac{1+1/x}{1-1/2}\right] = \beta\left[\frac{1}{x}+\frac{1}{x^3}+\frac{1}{x^5}+\cdots\infty\right)$$

$$= \ln\left[\frac{x+1}{x-1}\right] \quad \because x = \frac{1}{y^3-1}$$

$$\ln\left[\frac{\frac{1}{y^3-1}+1}{\frac{1}{y^3-1}-1}\right] = \ln\left[\frac{y^3}{2-y^3}\right]$$

**3. (b)** $y = 0.36 \log_{\frac{25}{100}}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots\right)$

$$y = 0.36\log_{\frac{1}{4}}\left(\frac{\frac{1}{3}}{1-\frac{1}{3}}\right) = 0.36\log_{2^{-2}}\left(2^{-1}\right)$$

$$y = 0.36\log_{2^{-2}}2^{-1} = \frac{0.36}{2}\log_2 2 = 0.4\log_2 2$$

$$y = 0.18$$

**4. (a)** $a>0, a\neq 1,\ x>0, x\neq 1$

$$\Rightarrow \frac{2}{\log_a x}+\frac{1}{\log_a x}+\frac{3}{\log_a a^2+\log_a x}=0$$

$$\Rightarrow \frac{2}{\log_a x}+\frac{1}{\log_a a+\log_a x}+\frac{3}{\log_a a^2+\log_a x}=0$$

$$\Rightarrow \frac{2}{\log_a x}+\frac{1}{1+\log_a x}+\frac{3}{2+\log_a x}=0$$

Let $\log_a x = t$

$$\Rightarrow \frac{2}{t}+\frac{1}{1+t}+\frac{3}{2+t}=0$$

$$\Rightarrow \frac{2(1+t)(2+t)+t(2+t)+3(1+t)t}{t(1+t)(2+t)}=0$$

$$\Rightarrow 6t^2+11t+4=0 \qquad \Rightarrow \qquad t=-\frac{1}{2},-\frac{4}{3}$$

So, it has 2 solutions.

**5. (a)** $|x+1| = \begin{cases} x+1 & x+1\geq 0 \Rightarrow x\geq -1 \\ -(x+1) & x+1<0 \Rightarrow x<-1 \end{cases}$

$$|2^x - 1| = \begin{cases} 2^x - 1 & 2x \ge 1 \Rightarrow x \ge 0 \\ -(2^x - 1)2^x < 1 \Rightarrow x < 0 \end{cases}$$

$$2^{|x+1|} - 2^x = |2^x - 1| + 1 \qquad [\text{As } x < -1]$$

$$\Rightarrow \quad 2^{-x-1} - 2^x = -2^x + 1 + 1 \Rightarrow 2^{-x-1} = 2$$

On equating

$$\Rightarrow \quad -x - 1 = 1 \Rightarrow x = -2$$

**6.** **(c)** $2^{2a} - 3(2^a.2^2) + 2^5 = 0$

Let $(2^a = t > 0) \Rightarrow t^2 - 3(4)t + 32 = 0$

$$\Rightarrow \quad t^2 - 12t + 32 = 0 \Rightarrow (t-4)(t-8) = 0$$

$$\Rightarrow \quad t = 4,8 \Rightarrow 2^a = 2^2, 2^3.$$

$\Rightarrow \quad a = 2,3 \Rightarrow$ two values

**7.** **(b)** $\begin{cases} (x+2)(x+4) > 0 \\ (x+2) > 0 \end{cases}$

hence, $x > -2$

$$\log_3 (x+2)(x+4) + \log_{3^{-1}} (x+2) < \frac{1}{2} \log_{3^{1/2}} 7$$

$$\Rightarrow \quad \log_3 (x+2)(x+4) - \log_3 (x+2) < \log_3 7$$

$$\Rightarrow \quad \log_3 (x+4) < \log_3 7 \Rightarrow x + 4 < 7 \Rightarrow x < 3.$$

Therefore, the solution set of inequality is $(-2,3)$.

**8.** **(c)** $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$

$$= \frac{1}{1 + \log_a bc} + \frac{1}{1 + \log_b ca} + \frac{1}{1 + \log_c ab}$$

$$= \frac{1}{\log_a a + \log_a bc} + \frac{1}{\log_b b + \log_b ca} + \frac{1}{\log_c c + \log_c ab}$$

$$= \frac{1}{\log_a abc} + \frac{1}{\log_b abc} + \frac{1}{\log_c abc}$$

$$= \log_{abc} a + \log_{abc} b + \log_{abc} c = \log_{abc} abc = 1$$

**9.** **(c)** Let $2^a = 3^b = 6^{-c} = k$

$\Rightarrow$ $2^a = k$, $3^b = k$, $6^{-c} = k$

$$\Rightarrow \quad 2 = k^{\frac{1}{a}}, 3 = k^{\frac{1}{b}}, 6 = k^{\frac{1}{c}} \Rightarrow 2 \times 3 = k^{\frac{1}{c}}$$

$$\Rightarrow \quad k^{\frac{1}{a}} \times k^{\frac{1}{b}} = k^{\frac{1}{c}}$$

$$\Rightarrow \quad \frac{1}{a} + \frac{1}{b} = -\frac{1}{c} \Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0 \Rightarrow \frac{ab + bc + ca}{abc} = 0$$

$\Rightarrow$ $ab + bc + ca = 0$

**10.** **(a)** $x = \frac{\log 5}{\log 3}$, $y = \frac{\log 25}{\log 17}$

$$y = \frac{2 \log 5}{\log 17} = \frac{2 \log 5}{2 \log \sqrt{17}} = \frac{\log 5}{\log \sqrt{17}}$$

As $\sqrt{17} > 3 \Rightarrow \log \sqrt{17} > \log 3$

$$\Rightarrow \quad \frac{1}{\log 3} > \frac{1}{\log \sqrt{17}} \Rightarrow \frac{\log 5}{\log 3} > \frac{\log 5}{\log \sqrt{17}}$$

$\Rightarrow$ $x > y$

**11.** **(c)** $\log_2 x = 10 \Rightarrow x = 2^{10}$

$$\log_x y = 100 \Rightarrow y = x^{100} \Rightarrow y = (2^{10})^{100} \Rightarrow y = 2^{1000}$$

**12.** **(c)** $(x_0, y_0)$ is solution of these two equations

$$(2x)^{In\,2} = (3y)^{In\,3}$$

Taking both sides of base e-

$$In(2x)^{In\,2} = In(3y)^{In\,3}$$

$$\Rightarrow \quad (\text{In } 2)(\text{In } 2 + \text{In } x) = (\text{In } 3)(\text{In } 3 + \text{In } y)$$

$$\Rightarrow \quad (\text{In } 2)^2 + \text{In } 2.\text{In } x = (\text{In } 3)^2 + \text{In } 3.\text{In } y \qquad (i)$$

$$3^{\text{In } x} = 2^{\text{In } y}$$

Taking log both sides of base e,

In $x$ In 3 = In $y$ In 2

$$\text{In } x = \frac{\text{In } y \text{ In } 2}{\text{In } 3} \qquad (ii)$$

Put this value in Eq. (*i*),

$$\text{In } 2\left(\text{In } 2 + \frac{\text{In } y \text{ In } 2}{\text{In } 3}\right) = \text{In } 3(\text{In } 3 + \text{In } y)$$

$$\text{In } y\left[\frac{(\text{In } 2)^2}{\text{In } 3} = -\text{In } 3\right] = (\text{In } 3)^2 - (\text{In } 2)^2$$

Hence, $\text{In } y = -(\text{In } 3) = \text{In } \frac{1}{3}$

Hence, $y = 1/3$

Put this value in Eq. (*ii*)

$$\text{In } x = \frac{\text{In } y \text{ In } 2}{\text{In } 3} = \frac{(-\text{In } 3)\text{In } 2}{\text{In } 3} = -\text{In } 2$$

Hence, $x = 1/2$.

**13.** **(a)** We have

$$a = \log_{12} 18 = \frac{\log_2 18}{\log_2 12} = \frac{\log_2 (2 \times 3^2)}{\log_2 (2^2 \times 3)} = \frac{\log_2 2 + 2\log_2 3}{\log_2 2^2 + \log_2 3}$$

$$a = \frac{1 + 2\log_2 3}{2 + \log_2 3}$$

$$b = log_{24} 54 = \frac{log_2 54}{log_2 24} = \frac{log_2(2 \times 3^3)}{log_2(2^3 \times 3)} = \frac{1 + 3\,log_2 3}{3 + log_2 3}$$

Let $\log_2 3 = t$

Hence, $ab + 5(a - b)$

$$= \left(\frac{1+2t}{2+t}\right)\left(\frac{1+3t}{3+t}\right) + 5\left(\frac{1+2t}{2+t} - \frac{1-2t}{3+t}\right) = \frac{t^2 - 5t + 6}{(t+2)(t+3)} = 1$$

**14.** **(d)** $x = n - 1,\ n \in N,\ y = n$ and $z = n + 1$

$$\log(1+xz) = \log(1+(n-1)(n+1))$$

$$= \log(1+(n^2-1)) = \log n^2 = 2\log n = 2\log y$$

**15. (a)** $x > 0,, x \neq 1, \frac{1}{2}, \frac{1}{4}$

$$\frac{\log_2 2 \log_2 2}{\log_2 x \log_2 2x} = \frac{\log_2 2}{\log_2 4x}$$

$$\Rightarrow \frac{1}{\log_2 x}\frac{1}{(1+\log_2 x)} = \frac{1}{(2+\log_2 x)}$$

$$\Rightarrow \frac{1}{t} \times \frac{1}{(1+t)} = \frac{1}{2+t}$$

$$\Rightarrow 2+t = t^2 + t \Rightarrow t^2 = 2$$

$$\Rightarrow t = \pm\sqrt{2} \Rightarrow \log_2 x = \sqrt{2} \text{ and } \log_2 x = -\sqrt{2}$$

$$\Rightarrow x = 2^{\sqrt{2}},\ 2^{-\sqrt{2}}$$

**16. (d)** Given, $3^x = 4^{x-1}$

Taking log both sides of base 3,

$$\log_3 3^x = \log_3 4^{x-1}$$

$$\Rightarrow x = (x-1)\log_3 4 \Rightarrow x = (x-1)\log_3 2^2$$

$$\Rightarrow x = 2(x-1)\log_3 2$$

$$\Rightarrow x = (2x-2)\log_3 2 \Rightarrow x = 2x\log_3 2 - 2\log_3 2$$

$$\Rightarrow 2\log_3 2 = 2x\log_3 2 - x$$

$$\Rightarrow 2\log_3 2 = x(2\log_3 2 - 1) \Rightarrow x = \frac{2\log_3 2}{(2\log_3 2 - 1)}$$

**17. (c)** $(2^x)^2 - 3(2^x.2^3) + 2^7 = 0$

Let $2^x = t > 0$

$$\Rightarrow t^2 - 3.2^3 t + 2^7 = 0 \Rightarrow t^2 - 24t + 128 = 0$$

$$\Rightarrow (t-16)(t-8) = 0 \Rightarrow t = 16, 8 \Rightarrow 2^x = 2^4, 2^3$$

$$\Rightarrow x = 4, 3 \Rightarrow \text{Sum of roots} = 4+3 = 7$$

**18. (b)** Let $x = \frac{1}{\sqrt{2}}\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 - \frac{1}{\sqrt{2}}\sqrt{1 - \frac{1}{\sqrt{2}}}}}$

$\Rightarrow x = \frac{1}{\sqrt{2}}(1-x) \Rightarrow x^2 = \frac{1}{2}(1-x)$, We know $x > 0$

$$\Rightarrow 2x^2 = 1 - x \Rightarrow 2x^2 + x - 1 = 0$$

$$\Rightarrow 2x^2 + 2x - x - 1 = 0 \Rightarrow (2x-1)(x+1) = 0 \Rightarrow x = \frac{1}{2}, -1$$

As $x > 0 \Rightarrow x = \frac{1}{2}$

$$6 + \log_{2^{-2}} 2^{-1} = 6 + \frac{1}{2}\log_2 2 = \frac{6}{1} + \frac{1}{2} = \frac{13}{2}$$

**19. (b)** Refer to solution 5.

**20. (b)** $3^3 \times 3 - \log_3 5 = 3^3 . 3^{\log_3 5^{-1}} = 3^3 \times 5^{-1} = \frac{27}{5}$

**21. (a)** $\log_e(1+x) = x - \frac{x^2}{3} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} - \ldots - \frac{x^{100}}{100} + \ldots$

$$In(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \frac{x^8}{4} + \ldots - \frac{(x^2)^{50}}{50}$$

Coefficient of $x^{100}$ in $In(1+x) - In(1+x^2)$

$$= -\frac{1}{100} + \frac{1}{50} = 0.01$$

**22. (a)**

Let x > 0. Put t = $log_5 x \Rightarrow x = 5^t$

$$x^{8log_5 x - 24} = (5^t)^{8t-24} = 5^{t(8t-24)} = 5^{-4}$$

So

$$8t^2 - 24t = -4 \Rightarrow 8t^2 - 24t + 4 = 0$$

$$\Rightarrow 2t^2 - t + 1 = 0$$

Roots:

$$t = \frac{6 \pm \sqrt{36-8}}{4} = \frac{3 \pm \sqrt{7}}{2}$$

Hence $x = 5^t = 5^{(3\pm\sqrt{7})/2}$

Product of all possible $x$:

$$x_1 x_2 = 5^{t_1 + t_2} = 5^3 = 125$$

**23. (c)** Given: $8^{x-1} = \left(\frac{1}{4}\right)^x$ rewrite as $(2^3)^{x-1} = (2^2)^{-x}$

$2^{3x-3} = 2^{-2x}$ equate exponents:

$3x - 3 = -2x \Rightarrow 5x = 3 \Rightarrow$ x $= \frac{3}{5}$.

Evaluate S = $\frac{1}{log_{x+1}4 - log_{x+1}5} + \frac{1}{log_{1-x}4 - log_{1-x}5}$

Substitute $x = \frac{3}{5}$, so $x + 1 = \frac{8}{5}$, $1 - x = \frac{2}{5}$

Using $log_b a = \frac{\log a}{\log b}$:

$$S = \frac{1}{\frac{\log - \log 5}{log\frac{8}{5}}} + \frac{1}{\frac{\log 4 - \log 5}{log\frac{2}{5}}} = \frac{log\frac{8}{5} + \log\frac{2}{5}}{\log 4 - \log 5} = \frac{log\frac{16}{25}}{\log 4 - \log 5} =$$

$$\frac{4\log 2 - 2\log 5}{2\log 2 - \log 5} = 2$$

**Ans.** 2